Unanswered Questions

Limits of knowledge, science, and the human journey in the contemporary world

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Limits of knowledge, science, and the human journey in the contemporary world

Title: Unanswered Questions.
Limits of knowledge, science, and human progress in the modern world.

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Executive Summary

The present work, titled **“Unanswered Questions,”** systematically examines the limits of knowledge, science, and the human trajectory in the modern world.
Its purpose is not to provide definitive answers, but to map the questions that remain open—either due to inherent theoretical constraints or due to social, institutional, and value-based tensions.

The central claim of the work is that unanswered questions do not represent a failure of knowledge or a sign of delayed progress.
On the contrary, they constitute a structural element of both scientific inquiry and social organization.
Understanding the nature of these questions is essential for shaping responsible political, technological, and institutional decisions.

Aim and Approach

The aim of the book is to clearly distinguish between:

- questions that have not yet been answered,
- questions that cannot be answered under the same terms, and
- questions that are systematically avoided due to political, social, or institutional cost.

The approach is interdisciplinary without conflating fields.
The work maintains clear boundaries between philosophical, social, and scientific analysis, while highlighting their points of convergence and interaction.
It is not proposed as an academic specialization manual, but as a conceptual and institutional framework for understanding.

Structure and Key Axes

The book is organized into three main parts:

A. Unanswered Questions of the Human Journey

This section examines fundamental issues of philosophy, social organization, and economics.
It analyzes concepts such as: progress and its limits; the relationship between freedom, responsibility, and power; democracy as institution or process; inequality and social cohesion; work and value in the 21st century.

The central conclusion is that many social questions remain unanswered not due to lack of knowledge, but because they involve conflicting values and collective choices that cannot be resolved technically.

B. Unanswered Scientific Questions

This section analyzes what constitutes a scientific question and how scientific knowledge is structured through theories, models, and hypotheses. It examines, indicatively:

- the fundamental open problems of the natural sciences,
- the origin of life and consciousness in biology,
- the limits of computability and artificial intelligence.

The work shows that unanswered scientific questions are not peripheral; they lie at the core of the most successful theories. Their existence indicates limits of method, not merely lack of data.

C. Science, Technology, and Society

The final part focuses on the encounter between scientific knowledge and social-political reality.
It highlights the relationship between science and political decision-making, the ethical dilemmas of technological power, and the notion of responsibility in knowledge.

The book argues that science cannot replace the democratic process, nor can technology alone determine social purposes.
Clarifying these boundaries is a prerequisite for institutional legitimacy.

Key Conclusions

From the overall analysis, several core conclusions emerge:

- Unanswered questions are necessary for the dynamism of knowledge and society.
- Confusion between ignorance and limits leads either to technocratic overconfidence or to skepticism.
- Scientific progress does not eliminate uncertainty; it transforms it.
- Technology increases power, not necessarily understanding.
- The responsibility of knowledge is an institutional—not merely individual—matter.

Institutional Value of the Work

The book can be used:

- as a framework for reflection by policymakers,
- as introductory material in interdisciplinary programs,
- as a basis for public dialogue on science, technology, and society.

It does not propose policy solutions or technical recipes.
Instead, it offers conceptual clarity, role distinction, and awareness of limits—elements essential for responsible decision-making in environments of uncertainty.

Final Assessment

The Executive Summary demonstrates that *"Unanswered Questions"* is not a collection of open problems but a coherent framework of thought.
Its central contribution lies in accepting uncertainty as an institutional and epistemic condition, rather than as a temporary deficiency.

In a world where the demand for quick and definitive answers intensifies, the work argues that responsible knowledge begins with recognizing its own limits.
This recognition constitutes its fundamental institutional value.

If you want, I can also refine the translation for a more academic tone, a more concise version, or a version suitable for publication.

Prologue

This work begins with a simple yet critical observation.

Despite the accumulation of knowledge, data, and technological capabilities, the modern world remains surrounded by questions that either have not been answered or cannot be answered in the way we typically formulate them.

These questions do not belong to a single field.

They cut across philosophy, society, economics, technology, and science.

Some concern the meaning and direction of the human journey, while others relate to the limits of scientific knowledge and our theoretical understanding.

This book approaches such questions not as weaknesses, but as structural features of human thought and organization.

Why Do Questions Remain Unanswered?

Unanswered questions do not exist because “we have not progressed enough.”

They exist because certain problems are multilayered, value-laden, or institutionally charged, and therefore do not admit single, definitive solutions.

On a social and philosophical level, many questions remain open because they:

1. involve conflicting values,
2. clash with dominant economic or political interests,
3. or require collective decisions that no technology can replace.

On a scientific level, questions remain unanswered not due to lack of effort, but because:

- existing theoretical frameworks are insufficient,
- competing theories coexist with different explanatory strengths,
- or the question itself touches the limits of the method.

This book does not treat unanswered questions as temporary gaps that will "inevitably be filled" in the future.

Instead, it views them as active points of tension through which knowledge, society, and science evolve.

The Difference Between Ignorance and Limit

A central claim of the work is that we must clearly distinguish ignorance from limit.

Ignorance concerns what we do not yet know:

insufficient data, inadequate tools, temporary theoretical weaknesses.

Science and research constantly work to reduce it.

Limit, by contrast, concerns what:

- cannot be fully described within a given theoretical framework,
- cannot be measured without altering its object,
- cannot receive a definitive answer without introducing value-based or philosophical assumptions.

Confusing ignorance with limit leads either to naïve optimism ("technology will solve everything") or to sterile skepticism.

This book proposes a third stance: the conscious acceptance of limits as a tool for understanding and responsibility.

Purpose and Direction of the Work

The purpose of the book is not to provide answers where none can exist, but to:

- map the types of unanswered questions,
- distinguish the social and philosophical from the strictly scientific,
- show how the limits of knowledge shape the human trajectory.

With this structure, *Unanswered Questions* is treated not as a catalogue of unresolved issues, but as a framework for understanding the conditions under which questions remain open and why this openness matters.

Introduction

Our era is defined by a paradoxical condition: never before has humanity possessed so much knowledge, so much data, and so many technological means; and yet, never before has it faced such fundamental and persistent questions about its future, its cohesion, and its direction.

The coexistence of this cognitive abundance with deep uncertainty is not a contradiction, but an indication that knowledge does not exhaust the field of understanding.

This book begins from the premise that unanswered questions are not merely temporary gaps awaiting completion. On the contrary, in many cases they constitute structural features of societies, sciences, and human thought. Their persistence does not signal failure; it reveals the boundaries within which human reasoning and action operate.

The purpose of this Introduction is to delineate the framework within which these questions will be examined: not as riddles to be solved, but as domains of inquiry that shape the very notion of knowledge and responsibility.

What We Mean by "Unanswered Questions"

The term *unanswered* is often used vaguely or superficially.

In the context of this work, it does not simply mean "we do not yet know."

It refers to questions that:

- do not admit a single, definitive answer,
- cannot be addressed without introducing value-based or philosophical assumptions,
- or change as the social and historical context changes.

In this sense, unanswered questions are not static.

They are reformulated, shifted, and reappear in new forms.

The question of progress, for example, remains open not because technological developments have been lacking, but because there is no universal agreement on what constitutes "progress" or who has the authority to define it.

The same applies to many scientific questions: the existence of multiple theories, the inability to unify frameworks, or the inherent limits of measurement are not mere technical obstacles, but indications of deeper theoretical boundaries.

The Dual Nature of Questions: Social and Scientific

The book is structured around a key distinction: that between unanswered questions of the human journey and unanswered scientific questions.

This distinction is not absolute, but functional.

The former concern issues of meaning, values, organization, and direction.

They involve philosophy, politics, sociology, economics, and technology.

They are not answered through experiments or equations, but through public dialogue, historical experience, and collective decision-making.

The latter concern the structure of knowledge, theoretical frameworks, and scientific methods.
Here, the problem is not the lack of data, but often the inadequacy of the conceptual tools with which we approach reality.
The existence of multiple theories for the same phenomenon is not a failure of science, but a sign that we are operating near its limits.

This distinction is crucial because it prevents a common confusion: the tendency to treat social or ethical questions as technical problems, or scientific impasses as mere ignorance that will automatically be resolved with more technology.

Science, Technology, and the Illusion of the Final Answer

One of the main motivations of this work is the need to challenge the widespread belief that scientific and technological progress inevitably leads to the gradual elimination of all questions.

Although appealing, this belief overlooks the fact that every new answer generates new questions—often more complex than the ones before.

Technology, in particular, is often presented as a neutral problem solving tool.

In reality, it embodies values, priorities, and constraints.

The questions it raises—about labor, privacy, autonomy, or power—cannot be answered through technical means alone.

This book seeks to show that recognizing limits is not an obstacle to progress, but a prerequisite for responsible action.
Accepting that certain questions will remain open does not lead to resignation, but to more mature forms of thought and decision-making.

How to Read This Book

This work is not offered as a manual of answers, but as a map of questions.

Each chapter can be read either independently or as part of a broader synthesis.

The reader is not asked to accept specific positions, but to become familiar with the distinctions, boundaries, and tensions that shape the contemporary world.

This Introduction serves as a foundation: it prepares the ground for a systematic exploration of unanswered questions, with respect both for knowledge and for its limits.

From this point onward, the book proceeds with its analysis, guided by the idea that the maturity of thought is not measured by the number of answers it provides, but by the quality of the questions it dares to pose.

PART A — The Unanswered Questions of the Human Journey

Chapter 1 — What “Unanswered” Really Means

The word *“unanswered”* is often used vaguely, almost automatically. It tends to imply a lack of knowledge, a delay, or an inability. In reality, however, the *unanswered* is not a single condition but a spectrum of different forms of uncertainty, each with its own meaning and consequences.

The purpose of this chapter is to clarify what we mean when we describe a question as unanswered. This clarification is crucial, because it determines whether a question is treated as a technical problem to be solved, a theoretical limit, or an issue that is consciously avoided. Without this distinction, any discussion about knowledge, science, or society remains confused.

1.1 Knowledge, Ignorance, and Uncertainty

Human knowledge does not develop linearly. It is not a simple accumulation of information that gradually leads to a complete understanding of the world. Instead, knowledge advances through successive revisions, errors, rejections, and reformulations. Within this process, ignorance is not merely the absence of knowledge but an integral part of understanding.

Ignorance concerns what we do not yet know but believe can become known. It is the domain of research, observation, and systematic study. Historically, much of scientific progress consists precisely in transforming ignorance into knowledge—from the structure of the atom to the functioning of the brain.

Uncertainty, however, is something different. It does not simply indicate a lack of information but instability within the very framework of understanding. It may arise from conflicting data, inadequate theoretical models, or the inability to define fundamental concepts. In such cases, increasing the amount of data does not necessarily lead to clarity; it often leads to greater complexity.

Confusing knowledge, ignorance, and uncertainty leads to false expectations. When every unanswered question is treated as mere ignorance, the belief is cultivated that time and technology will inevitably provide all answers. This work challenges that assumption.

1.2 Questions Without a Definitive Answer

Some questions remain unanswered not because information is lacking, but because they do not possess a definitive answer by nature. These are questions that touch on concepts such as meaning, value, justice, or progress. Their answers depend on historical, cultural, and philosophical contexts that constantly change.

For example, the question *"What constitutes progress?"* cannot be answered in a single, universal way. Different societies, eras, and groups provide different—and often conflicting—answers. The same applies to questions concerning social organization, the relationship between freedom and security, or the role of technology in human life.

Even in science, there are questions that do not admit a definitive answer in the sense of a complete and irreversible solution. The existence of multiple theories for the same phenomenon, the inability to unify different frameworks, or the inherent limits of measurement show that scientific knowledge does not always converge on final answers but on provisional and functional ones.

Accepting that some questions have no definitive answer does not imply abandoning the effort to understand. On the contrary, it allows for a more balanced and mature stance toward knowledge, recognizing its limits without confusing them with failure.

1.3 Questions That Are Avoided

A distinct category of unanswered questions consists of those that remain unanswered not because they cannot be answered, but because we choose not to answer them. These are questions that challenge established interests, question dominant narratives, or require decisions with political and moral cost.

At the social level, such questions often concern the distribution of power, the modes of producing and distributing wealth, or the boundaries of institutional legitimacy. Their avoidance is not accidental; it is a mechanism for maintaining stability—even when that stability is accompanied by inequalities or dead ends.

In science and technology, avoided questions often relate to the social consequences of knowledge. The issue is not only what we can do technically, but whether and how we should do it. When technical capability advances faster than public discussion, ethical questions tend to be postponed or downplayed.

Recognizing the questions that are avoided is essential for any serious approach to the unanswered. In these cases, the problem is not epistemic but institutional and political. Silence does not indicate ignorance—it indicates choice.

Transitional Note

The distinction between ignorance, uncertainty, the absence of a definitive answer, and avoidance forms the foundation upon which the following chapters will build. Only by understanding what kind of *unanswered* we are dealing with in each case can we approach, with seriousness, both the questions of the human journey and the limits of scientific knowledge.

Chapter 2 — Philosophy and Meaning

Philosophy is not merely the historical background of the sciences or an abstract engagement with concepts.

It is the domain in which the fundamental questions about the meaning, direction, and value of human existence are posed explicitly, without the expectation of definitive resolution.

In this sense, many of the unanswered questions that concern contemporary society are not new; they reappear with different intensity and under new conditions.

This chapter examines three central themes: the concept of progress, the relationship between freedom and responsibility, and the question of meaning in a world increasingly mediated by technology.

2.1 Is There Progress, or Only Change?

The idea of progress is one of the strongest pillars of modern thought.

It has been associated with scientific discovery, industrial development, and the expansion of individual rights.

However, the concept of progress is often conflated with the concept of change, as if every transformation necessarily implies improvement.

The question of whether there is progress or merely change remains unanswered because it presupposes a criterion of evaluation.

Progress in relation to what?

Economic prosperity, technological power, social justice, or human well-being?

Different answers lead to different—and often conflicting—conclusions.

Historically, there are periods in which technological progress coexists with social regression or moral crisis.

Increased productivity does not always reduce inequality, nor does scientific knowledge guarantee wiser collective decisions.

This makes the very notion of progress deeply contested.

Philosophy does not offer a final answer to the question, but it highlights its fundamental difficulty: without clearly articulated goals and values, change cannot be classified as either progress or decline.

The unanswered here is not a sign of failure but a reminder that the direction of the human journey is not self-evident.

2.2 Freedom, Responsibility, and Power

Freedom is a central value of modern societies, yet its meaning is far less clear than often assumed.

Freedom from what, and freedom for what?

Individual autonomy, collective self-determination, or simply the absence of constraints?

The question becomes more complex when responsibility is introduced.

Freedom without responsibility becomes arbitrariness, while responsibility without freedom becomes coercion.

The balance between the two is not given; it is the subject of ongoing social negotiation.

Power intervenes precisely at this point.

Every society organizes mechanisms that regulate behavior, set boundaries, and distribute responsibilities.

The unanswered question concerns when the exercise of power protects freedom and when it restricts it.

There is no universal point of equilibrium; it shifts according to historical and cultural context.

In the contemporary world, the issue becomes even more complex, as power is exercised not only through institutions but also through technological systems, algorithms, and infrastructures.

Responsibility becomes diffused, decision-making moves away from the individual, and freedom is redefined within environments not fully controlled by their users.

The question of the relationship between freedom, responsibility, and power remains unanswered not because theories have not been formulated, but because no theory can encompass all conditions and all consequences of human action.

2.3 Meaning in a Technological World

Technology is no longer merely a tool.
It shapes the way we work, communicate, perceive time, and understand ourselves.
In this environment, the question of meaning acquires a new dimension.

Traditionally, meaning was drawn from religion, community, work, or participation in collective endeavors.
In the technological world, many of these sources are weakened or transformed.

Work becomes automated, social relations become digitized, and experience becomes fragmented.

The unanswered question is not whether technology "removes" meaning, but whether and how it can support it.
Can a world of high efficiency and speed offer depth of experience?

Can constant informational availability coexist with reflection and silence?

Philosophical discussion does not arrive at a definitive answer.

It does, however, indicate that meaning is not produced automatically by technical progress.

It requires conscious stance, choices, and boundaries.

Without these, technology risks functioning as a substitute for meaning, offering activity instead of purpose.

Transitional Note

The questions examined in this chapter cannot be resolved through more information or better tools.
They belong to the core of philosophical inquiry and accompany every historical phase of the human journey.
Their persistence is not a weakness of thought but an indication that meaning, freedom, and progress are not fixed quantities—they are open fields of responsibility.

In the next chapter, the analysis shifts from the philosophical background to collective organization and social structures, where unanswered questions acquire institutional and practical form.

Chapter 3 — Society and Organization

Society is not merely the sum of the individuals who compose it.

It is a system of institutions, rules, practices, and narratives that organize collective life and determine what is considered acceptable, just, or possible.

The unanswered questions concerning social organization do not arise from theoretical vagueness but from the tension between ideals and realities.

This chapter examines three fundamental axes of contemporary society: the nature of democracy, the relationship between inequality and cohesion, and the transformation of society under the weight of information and technology.

In all three cases, the questions remain unanswered not because solutions have not been proposed, but because no solution is neutral or definitive.

3.1 Democracy: Institution or Process?

Democracy is often presented as a given institution: a set of rules, electoral procedures, and constitutional guarantees that are assumed to secure popular sovereignty.

However, experience shows that the mere existence of institutions does not guarantee democratic functioning.

The unanswered question is whether democracy should be understood primarily as a stable institution or as an ongoing process.

If it is an institution, emphasis is placed on institutional safeguards, legality, and continuity.
If it is a process, democracy requires continuous participation, contestation, and renewal.

This distinction is not merely academic.

In societies where democracy is reduced to compliance with formal procedures, citizens often experience alienation, declining trust, and rising cynicism.

Conversely, understanding democracy as a living process entails conflict, uncertainty, and at times instability.

The question remains unanswered because every society must balance stability and participation.

There is no universal model that resolves this tension.

Democracy is not something that is “possessed”; it is something that is practiced, and its practice is always incomplete.

3.2 Inequality and Social Cohesion

Inequality is a persistent feature of societies, but in the contemporary world it takes on new forms and intensities.

Economic, educational, digital, and institutional inequalities coexist and reinforce one another, creating fractures in collective cohesion.

The unanswered question is not whether inequality exists, but which forms of inequality are tolerable and by what criteria.

Complete equality has never been historically achievable, while full acceptance of inequality undermines social cohesion and democratic legitimacy.

Societies attempt to manage this tension through redistribution, social policies, and mechanisms of inclusion.
However, every such intervention involves choices of values and priorities.

Economic efficiency often clashes with social justice, and the notion of “opportunity” does not always align with the notion of “outcome.”

Social cohesion does not arise automatically from reducing inequalities but from a shared sense of participation and fairness.

When large segments of society feel excluded or invisible, institutions lose legitimacy—even if they function correctly in formal terms.

The question of the relationship between inequality and cohesion remains unanswered because it concerns not only numbers and indicators but the very conception of "we" within a society.

3.3 Information Society or Society of Control?

Digital technology promises access to knowledge, transparency, and individual empowerment.

The idea of the "information society" rests on the belief that the diffusion of information leads to more informed citizens and more open institutions.

At the same time, however, a different face of the same reality emerges: the systematic collection of data, the monitoring of behaviors, and the algorithmic management of social life.

The question is whether contemporary society is moving toward empowerment or toward control—or whether these two directions inevitably coexist.

Information is not neutral.

Its selection, classification, and interpretation determine what becomes visible and what remains invisible.

In this context, knowledge can function both as a means of emancipation and as a tool of discipline.

The unanswered question concerns not only technology but its institutional and social integration.

Who controls the data?

By what criteria are decisions made?

How is accountability ensured in systems that operate beyond immediate human understanding?

The information society risks becoming a society of control when technical efficiency outweighs democratic transparency.

The question remains open because the balance between innovation and freedom has no predetermined solution.

Transitional Note

The social questions examined in this chapter show that the organization of collective life is not a technical problem but a continuous process of negotiation.

Democracy, social cohesion, and information are not fixed states but dynamic fields of tension.

In the next chapter, the analysis shifts to the economy and labor, where unanswered questions take on material and everyday form, directly shaping the ways in which we live and survive.

Chapter 4 — Economy and Work

The economy and work constitute the material foundation of social life.

Through them, not only the conditions of survival are determined, but also the possibilities for participation, dignity, and social recognition.

The unanswered questions that concern them are not abstract; they manifest daily in working conditions, in the distribution of wealth, and in the perception of what is considered socially useful.

This chapter examines three central tensions: the limits of growth, the transformation of work in the 21st century, and the relationship between value, money, and social usefulness.

In all three cases, the unanswered does not stem from a lack of economic models but from the inability to agree on goals.

4.1 Growth Without Limits?

Economic growth is often presented as a self-evident goal.

Increases in GDP, productivity, and consumption are treated as indicators of success, regardless of social or environmental consequences.

The question of whether growth can—or should—be unlimited remains open because it touches on fundamental assumptions about the role of the economy.

On one hand, growth is associated with poverty reduction, improved infrastructure, and expanded opportunities.

On the other hand, the unlimited expansion of production and consumption clashes with natural resources, climate stability, and social cohesion.

The unanswered question is not only whether there are physical limits to growth but whether there are conceptual limits.

At what point does increased economic activity cease to improve quality of life?

When does it become an end in itself, serving indicators rather than human needs?

Different economic schools offer competing answers, but none has resolved the tension between growth, sustainability, and justice.

The persistence of the question shows that the problem is not technical but value-based.

4.2 Work in the 21st Century

Work has historically been a central element of identity and social integration.

In the 21st century, however, the nature of work is rapidly transforming.

Automation, digitization, and flexible forms of employment are redefining what it means to work.

The unanswered question is whether work will remain the primary mechanism of social integration or whether we are moving toward a society in which employment is sporadic, unstable, or unevenly distributed.

Increases in productivity do not necessarily lead to reduced working hours or improved conditions.

At the same time, work is increasingly detached from security.

Temporary contracts, gig-based labor, and algorithmic evaluation create new forms of dependency and uncertainty.

The question is not only how jobs will be created but what kind of work is considered dignified and socially acceptable.

The debate about the future of work remains unanswered because it requires decisions about the relationship between technological capability, social protection, and human dignity.

No technology can make these decisions automatically.

4.3 Value, Money, and Social Usefulness

At the core of economic thought lies the question of value.

What makes something valuable?

Is it the market, labor, scarcity, or social usefulness?

Money functions as a general mediator, but it is not identical to value.

The unanswered question arises from the gap between what is paid and what is socially necessary.

Work that sustains collective life—care, education, social cohesion—is often economically undervalued, while activities with questionable social contribution receive high rewards.

This contradiction cannot be resolved solely through market mechanisms.

It concerns deeper choices about what is recognized, rewarded, and considered essential for the functioning of society.

Money, as a measure of value, simplifies complexity but also conceals it.

The question of the relationship between value, money, and social usefulness remains unanswered because there is no neutral system of evaluation.

Every economic arrangement embodies specific assumptions about what matters.

Transitional Note

Economy and work are fields where unanswered questions have immediate and tangible consequences.

Growth, labor, and value are not merely technical concepts but carriers of social choices and conflicts.

In the next chapter, the analysis shifts to what constitutes a scientific question and how the limits of knowledge appear in a different—but equally decisive—form within the domain of science.

PART B — The Unanswered Scientific Questions

Chapter 5 — What Counts as a Scientific Question

The transition from questions about society and the human journey to questions about science requires a crucial clarification: not every question can become a scientific question, and not every scientific question can be answered in the same way.
Science is not merely a body of knowledge but a specific mode of questioning, governed by rules, constraints, and methodological commitments.

This chapter aims to define what makes a question scientific, how scientific knowledge is organized through theories and models, and when a question ceases to be scientifically answerable without losing its significance.

5.1 Theory, Model, Hypothesis

In scientific practice, the terms *theory*, *model*, and *hypothesis* are often used interchangeably in everyday language.

In reality, however, they correspond to different levels of knowledge and explanation.

A **hypothesis** is a specific, testable proposition about the relationship between phenomena.
It is formulated in a way that allows confirmation or rejection through observation or experiment.

A hypothesis is the starting point of scientific inquiry, but it does not constitute knowledge on its own.

A **model** is a simplified representation of reality.

Its purpose is not to describe everything but to illuminate particular aspects of a phenomenon.

Models incorporate assumptions, reduce complexity, and allow predictions within defined limits.

Their usefulness depends not on their "truth" but on their explanatory and predictive power.

A **theory** is the most complex level.

It unifies multiple models and hypotheses into a coherent framework, offering a generalized interpretation of phenomena.

A scientific theory is not a guess; it is the result of long-term research, continuous testing, and revision.

A scientific question is always situated in relation to these levels.

It does not float in a vacuum; it emerges from the limits, inconsistencies, or gaps of an existing theoretical framework.

5.2 The Notion of Scientific Completeness

We often speak of "complete" or "incomplete" scientific knowledge. Yet the notion of scientific completeness is more complex than it appears. It does not mean knowing everything, nor does it imply that a theory explains every possible aspect of reality.

Scientific completeness means that within a given framework:

- the fundamental concepts are clearly defined,
- the relationships between them are coherent,
- and the resulting explanations or predictions are testable.

A theory can be scientifically complete and still limited.

It may apply only to specific scales, conditions, or levels of analysis.

When extended beyond these limits, contradictions and unanswered questions emerge.

The existence of multiple theories in a scientific field is not a sign of confusion but often a sign of maturity and productive tension.

Competing theories coexist when available data allow more than one consistent interpretation.

Science does not choose theories based solely on aesthetics or simplicity but on overall performance.

The unanswered scientific question often arises precisely at these points: where theories are adequate but not unified; powerful but not universal.

5.3 When a Question Cannot Be Answered Scientifically

A critical yet often overlooked issue is that some questions cannot be answered scientifically—not because they lack meaning, but because they exceed the limits of the scientific method.

A question ceases to be scientifically answerable when:

- it cannot be formulated in testable terms,
- it does not allow for falsification,
- or it requires value judgments that cannot be derived from empirical data.

Questions about the meaning of life, the moral value of an action, or the “ultimate purpose” of nature are not scientific questions, even though they are absolutely fundamental.
Transferring them into the scientific domain often leads not to definitive answers but to the emergence of new, narrower, and often unanswered questions.

Even within science, there are limits that do not stem from ignorance but from the structure of knowledge itself.

The impossibility of fully describing certain phenomena simultaneously, theoretical dead ends, and methodological constraints are not temporary obstacles but indications of boundaries.

Recognizing these limits does not diminish science; it protects it from uncritical expansion beyond its methodological scope and from being transformed from a method of inquiry into a system of universal answers.

Transitional Note

Understanding what constitutes a scientific question is essential for approaching the unanswered scientific issues that follow.

Only by distinguishing levels of knowledge, the limits of theories, and the capacities of method can we meaningfully discuss the open problems of science.

In the next chapter, the analysis turns to the natural sciences, where unanswered questions are not merely theoretical but touch the fundamental structure of reality.

Chapter 6 — The Natural Sciences

The natural sciences are often presented as the domain where questions receive definitive answers.

Mathematically formalized theories, precise measurements, and verified experiments create the impression of an almost complete picture of nature.

Yet at the core of modern physics one encounters some of the most persistent unanswered questions.

These questions are not peripheral.

They concern the composition of the universe, the compatibility of fundamental theories, and the very limits of observation and measurement.

Their existence does not weaken physics; on the contrary, it reveals the points at which knowledge reaches its boundaries.

6.1 Dark Matter and Dark Energy

One of the most striking unanswered questions in modern physics is that most of the universe appears to consist of something we do not know.

Observations of galactic motion and the expansion of the universe cannot be adequately explained by known forms of matter and energy.

The concept of **dark matter** is introduced to explain gravitational phenomena that do not align with visible matter.

Similarly, **dark energy** is proposed to account for the accelerating expansion of the universe.

In both cases, these concepts function as theoretical entities that bring coherence to observations, without direct empirical confirmation.

The unanswered question is not simply *what* dark matter and dark energy are, but whether these concepts correspond to real physical phenomena or whether they fill gaps in deeper theoretical frameworks.

It is always possible that the observed anomalies do not require new forms of matter but a revision of the very laws we consider fundamental.

The persistence of the problem shows that physics, even in its most successful models, has not completed its description of reality on a cosmic scale.

6.2 Unifying Physical Theories

Modern physics rests on two extraordinarily successful yet incompatible theoretical frameworks.

On one side, the theories describing fundamental particles and their interactions.

On the other, the theory describing gravity and the structure of spacetime on large scales.

Both frameworks have been confirmed with remarkable precision within their domains.
However, when they are applied simultaneously—in extreme conditions such as the early universe or the interior of black holes—contradictions arise.

The unanswered question of unification concerns not only the mathematical compatibility of the theories but the deeper understanding of what is fundamental in nature.

Is space and time continuous or discrete?

Are particles basic entities or manifestations of deeper structures?

The difficulty of unification suggests that existing theories, however successful, may be approximations rather than final descriptions.

The question remains open not due to lack of effort but because it may require a radically new way of thinking.

6.3 The Limits of Measurement

Scientific knowledge depends on measurement. Without the ability to observe and quantify, a phenomenon can hardly be incorporated into the scientific domain.

However, measurement is not a neutral process.

It has limits—both technical and fundamental.

In some cases, measurement limits arise from technological constraints.

The history of physics shows that improved instruments often lead to new discoveries.

But there are also limits that cannot be overcome simply with better technology.

At microscopic scales, the act of measurement itself affects the system being measured.

At cosmic scales, observation is limited by the finite speed of information and the age of the universe.

We cannot observe everything, nor can we measure with unlimited precision.

The unanswered question here is not how to measure better, but what knowledge means when complete measurement is inherently impossible.

Physics must operate with partial, indirect, and statistical descriptions of reality.

Transitional Note

The unanswered questions of the natural sciences show that even the most precise scientific fields encounter limits that cannot be easily bypassed. Dark matter, the unification of theories, and the limits of measurement are not merely technical problems to be solved but points where knowledge touches the unknown.

In the next chapter, attention shifts to the biological sciences, where unanswered questions take on a different character, tied to the complexity of life and consciousness.

Chapter 7 — Biology and Life

Biology occupies an intermediate position between the natural sciences and the sciences of complexity.

It studies phenomena that obey physical laws yet exhibit properties not easily reducible to their constituent parts.

Life is not merely matter in motion; it is organization, history, and function.

The unanswered questions of biology concern the origin of life, the emergence of consciousness, and the dynamics of evolution.
In all these cases, the problem is not only what we do not know but how the question must be formulated in order to be scientifically fruitful.

7.1 The Origin of Life

The question of the origin of life is one of the most fundamental and persistent problems in science.

How does a self-sustaining, self-replicating, and evolving system arise from non-living matter?

Despite significant advances in chemistry, molecular biology, and geoscience, the answer remains open.

There are scenarios describing possible mechanisms: self-organizing chemical structures, molecules with catalytic properties, proto-cellular forms.

However, no theory has fully bridged the gap between chemical complexity and biological functionality.

The unanswered question concerns not only when or where life emerged but how it acquires properties such as metabolism, reproduction, and evolutionary dynamics.
The transition from chemistry to biology is not merely quantitative; it is qualitative.

The difficulty of the problem shows that life cannot be fully described as the sum of molecules.
It requires concepts such as organization, information, and function—concepts not easily reducible to physicochemical terms.

7.2 Consciousness and the Brain

Consciousness is perhaps the most enigmatic phenomenon studied by biology and related sciences.

Despite detailed knowledge of the brain's structure and function, the experience of consciousness—the sense of "I am"—remains insufficiently explained.

The unanswered question is how subjective experience arises from neural processes.
We can map brain regions, measure electrical activity, and predict behaviors, but the connection between these data and lived experience remains unclear.

Some approaches treat consciousness as an emergent property of complex neural systems.
Others attempt to link it to specific forms of information processing or integration.

Despite the variety of theories, none has achieved broad acceptance or full explanatory power.

The difficulty is not only empirical but conceptual.

Consciousness cannot be directly observed from a third-person perspective; it is lived.
This places limits on objective measurement and makes the problem unique within science.

7.3 Evolution and Complexity

Evolutionary theory is one of biology's most powerful explanatory frameworks.

It accounts for the diversity of life through mechanisms such as natural selection and heredity.

However, evolution alone does not fully explain the emergence of increasing complexity.

The unanswered question concerns how and under what conditions complex structures and functions arise.

Natural selection explains adaptation but does not always adequately describe the transition from simple to highly complex systems.

Biological complexity is not linear.

It develops through historical pathways, random events, and interactions across multiple levels.

Genes, cells, organisms, and ecosystems form networks that cannot be fully analyzed through reductionist methods.

The question remains unanswered because it requires a synthesis of evolutionary theory, systems thinking, and mathematical tools that are still under development.

Complexity is not merely the result of more components but of relationships and history.

Transitional Note

The unanswered questions of biology show that life is not fully accessible through a single method or theory.
The origin of life, consciousness, and complexity mark points where science encounters its limits without losing its legitimacy.

In the next chapter, the analysis shifts to technology and computation, where unanswered questions concern not only what we can build but what we can understand and control.

Chapter 8 — Technology and Computation

Technology is perhaps the most visible and immediate outcome of scientific knowledge.
Computation, as a general form of standardized information processing, lies at the core of most modern technologies.

However, the impressive progress of computational systems is often mistaken for progress in understanding.

The unanswered questions concerning technology and computation relate not only to what we can build but to what we can understand, explain, and control.

This chapter examines three central themes: the relationship between artificial intelligence and understanding, the fundamental limits of computability, and the possibility of technology advancing ahead of its own theory.

8.1 Artificial Intelligence and Understanding

Artificial intelligence is often presented as evidence that machines can perform functions once considered exclusively human.

Systems that recognize patterns, generate language, or make decisions reinforce the impression that understanding can be reduced to computational processes.

The unanswered question is whether successful performance of a function implies understanding.
A system may produce correct answers without possessing awareness of the meaning of what it processes.

The distinction between behavioral competence and conceptual understanding remains crucial.

The difficulty lies in the fact that understanding is not easily defined in functional terms.

It involves intention, context, meaning, and often a lived experiential dimension.

Artificial intelligence can simulate aspects of human behavior, but it is unclear whether and how these simulations correspond to understanding.

The question remains open not because computational methods are insufficient, but because we do not know whether understanding is a property that can be fully described computationally or whether it requires a different conceptual framework.

8.2 The Limits of Computability

Computation is often treated as a universal problem-solving tool.

The increase in computational power creates the expectation that every problem is, sooner or later, computable.

However, the theory of computation itself sets clear limits on this view.

There are problems that no algorithmic system can solve, regardless of resources or time.

Other problems are theoretically computable but practically inaccessible due to exponential complexity.

These limits are not technical obstacles; they are fundamental features of the concept of computation.

The unanswered question concerns not only which problems are computable but what this means for knowledge.

If certain questions cannot be solved algorithmically, then society's reliance on computational systems has inherent boundaries.

Recognizing the limits of computability undermines the idea of technological omnipotence.
It shows that the computational approach is powerful but not universal, and that some forms of understanding or decision-making cannot be fully automated.

8.3 Technology Without Theory?

Historically, technology often developed alongside theoretical understanding.

In the modern era, however, a reversal is increasingly observed: high-performance technological systems precede their full theoretical interpretation.

Especially in complex computational systems, functionality does not always come with transparency.

Systems “work,” but it is not always clear why.

The absence of interpretability raises questions about reliability, control, and accountability.

The unanswered question is whether sustainable technology can exist without adequate theoretical grounding.

History shows that lack of understanding does not prevent short-term success but creates long-term risks.

When systems fail or produce undesirable consequences, the absence of theory makes correction difficult.

Technology without theory turns knowledge into practical capability without conceptual foundation.

This does not negate its value but limits the possibility of responsible use.

Transitional Note

The unanswered questions of technology and computation show that increased power does not equate to increased understanding.

Artificial intelligence, the limits of computability, and the gap between functionality and theory reveal that technology cannot replace reflective thought.

In the next chapter, the analysis turns to the encounter between science and society, where unanswered questions acquire political, ethical, and institutional dimensions.

PART C — Boundaries and Syntheses

Chapter 9 — When Science Meets Society

Science does not develop in a social vacuum.

The questions that are posed, the research that is funded, and the applications that are implemented are influenced by political priorities, economic interests, and cultural values.

When scientific knowledge leaves the laboratory and becomes embedded in social practice, it turns into a force with real consequences.

The unanswered questions in this domain concern not only what we know, but how what we know is used and who bears responsibility for its consequences.

This chapter examines three critical axes of this encounter: the relationship between science and politics, the tension between technological capability and ethics, and the notion of responsibility in knowledge.

9.1 Scientific Knowledge and Politics

Scientific knowledge is often presented as a neutral basis for political decision-making.
In practice, however, the relationship between science and politics is complex and bidirectional.
Science can inform politics, but it cannot replace it.

The unanswered question concerns how scientific knowledge is translated into political action.

Scientific data describe states of affairs and probabilities; they do not determine goals or values.

Choosing political measures requires value judgments, priorities, and acceptance of costs—elements that do not arise from the scientific method.

Often, science is invoked to legitimize political decisions that have already been made.

In other cases, it is ignored when its conclusions conflict with dominant interests or short-term strategies.

In both situations, scientific knowledge becomes instrumentalized.

The question remains unanswered because there is no clear boundary between the advisory role of science and the political responsibility of decision-making.

Maintaining this distinction is essential for democratic legitimacy and scientific credibility.

9.2 Technology and Ethics

Technology expands human capabilities but simultaneously multiplies the consequences of our choices.

Every new technical possibility raises the question not only of whether it *can* be applied but whether it *should*.

The unanswered question of the ethics of technology cannot be resolved through technical criteria.

Efficiency, innovation, and competitiveness do not equate to moral acceptability.

On the contrary, they often conflict with values such as privacy, equality, and human dignity.

The difficulty is intensified by the speed of technological development.

Ethical and institutional structures move more slowly than technical capabilities, creating a gap in regulation and understanding.

The result is the reactive management of consequences that could have been anticipated.

The question remains open because there is no universal ethical framework applicable to all technologies and all social contexts.

The ethics of technology requires ongoing public dialogue and revision, not ready-made answers.

9.3 The Responsibility of Knowledge

Knowledge is not merely the acquisition of information; it is power.

As our ability to intervene in nature, society, and the human being increases, the question of responsibility becomes more pressing.

The unanswered question here concerns who bears responsibility for the use of knowledge.
Is it the scientist who produces it, the institution that funds it, the policymaker who legislates it, or the society that accepts it?

Responsibility becomes diffused, making accountability difficult.

At the same time, appeals to the "neutrality of science" can function as a denial of responsibility.

If knowledge is treated merely as a tool, then consequences are attributed solely to its use.

Yet the choice of what is researched and how is not neutral.

The responsibility of knowledge does not imply restricting research but recognizing its consequences.

The question remains unanswered because it concerns the relationship between the freedom of knowledge and social responsibility—a relationship that cannot be fully regulated in advance.

Transitional Note

The encounter between science and society reveals that unanswered questions arise not only at the boundaries of knowledge but also at the boundaries of collective responsibility.

Scientific progress without social reflection risks losing its legitimacy, while social decision-making without scientific understanding risks becoming arbitrary.

In the next and final chapter, the book turns to the overall synthesis: how we live, decide, and move forward in a world where unanswered questions are not eliminated but accompany us.

Chapter 10 — Living with the Unanswered

This book does not lead to a final answer, nor does it seek to close the questions it has brought to light.

On the contrary, it concludes precisely where thought must remain open.

Living with the unanswered does not mean abandoning knowledge or action; it means accepting a fundamental condition of human existence.

Uncertainty, the limits of science, social tensions, and ethical dilemmas are not temporary anomalies that will disappear with further progress.

They form the permanent framework within which decisions are made, institutions are built, and meaning is shaped.

10.1 Uncertainty as a Condition

The modern era is often described as an age of crises—economic, environmental, technological, institutional.

Yet behind these crises lies a deeper reality: uncertainty is not the exception but the rule.

Science allows us to predict, calculate, and control aspects of the world with unprecedented precision.

At the same time, it reveals ever more limits—theoretical, methodological, and practical.

Society, for its part, must operate in environments where the consequences of choices cannot be fully foreseen.

The unanswered is not merely a deficit of knowledge; it is a condition of decision-making.
Every significant human choice—individual or collective—is made without complete information and without guaranteed outcomes.

The demand for absolute certainty leads either to paralysis or to illusions of control.

Accepting uncertainty does not mean fatalism.

On the contrary, it allows for a more realistic stance toward knowledge and responsibility.

When we recognize that we cannot know everything, we become more careful in our decisions and more open to correction and revision.

10.2 Why the Unanswered Is Necessary

In a world that rewards quick answers, certainty, and efficiency, unanswered questions are often treated as weaknesses.

This book argues the opposite: the unanswered is necessary.

Without unanswered questions, thought stagnates.

Science becomes technical routine, politics becomes administration without vision, and technology becomes an end in itself.

Unanswered questions function as points of tension that keep inquiry, disagreement, and creativity alive.

Moreover, unanswered questions act as boundaries against the arrogance of omniscience.
They remind us that human knowledge is historical, situated, and finite.

This awareness does not diminish the value of knowledge; it makes it more responsible.

Finally, the unanswered is necessary for meaning.

Meaning does not arise from the complete resolution of all problems but from our relationship to what remains unresolved—from the way we choose to live, decide, and coexist within an imperfect and open world.

Concluding Reflection

Unanswered questions are not the end of thought but its beginning.

They are not failures of knowledge but the space where knowledge meets responsibility, ethics, and freedom.

Living with the unanswered does not mean abandoning the search for answers.

It means learning to recognize when an answer is temporary, when it is insufficient, and when it should not be given.

In this recognition lies perhaps the most mature form of human understanding.

Epilogue

This work has traversed a wide spectrum of unanswered questions: from philosophy and social organization to the natural sciences, biology, technology, and computation.

Its purpose was never to exhaust these topics nor to provide definitive answers.

Rather, it sought to systematically map the boundaries within which contemporary knowledge is formed and collective decisions are made.

One of the central conclusions that emerges is that unanswered questions are not a pathology of knowledge but one of its structural features.

Science advances precisely because it identifies the points at which existing theories, models, and methods are insufficient.

Society, likewise, evolves through the continuous negotiation of values, aims, and limits—not through the elimination of uncertainty.

The distinction between questions that have not yet been answered and those that cannot be answered under the same terms proved crucial.

Confusing these categories leads either to an overestimation of technological and scientific power or to an unjustified dismissal of knowledge.

This work argues that intellectual maturity lies in the ability to discern limits, not in denying them.

Particular emphasis was placed on the encounter between science and society.

Here, unanswered questions acquire institutional, political, and ethical weight.

Scientific knowledge can inform decisions, but it cannot replace democratic responsibility.
Technology can expand the scope of action, but it cannot determine its purposes.

Shifting these decisions to technical systems or expert bodies does not eliminate the questions; it merely renders them less visible.

Within this context, the responsibility of knowledge emerges as a central issue.

The production, dissemination, and application of knowledge are not neutral processes.
They embody priorities, exclusions, and consequences that require institutional accountability and social oversight.

Invoking the neutrality of science or technology does not absolve responsibility; it often obscures it.

This Epilogue does not aim to close the discussion.

Its purpose is to make clear that living alongside unanswered questions is not a weakness of the modern world but a prerequisite for institutional and epistemic maturity.
Societies that recognize their limits are better able to plan with foresight, correct their mistakes, and avoid turning knowledge into an instrument of arbitrary power.

Ultimately, the question is not whether the unanswered will disappear, but how we will confront them.

Whether we will attempt to cover them with simplistic certainties or consciously integrate them into the way we think, decide, and act.

The latter path is more demanding, but also more compatible with the values of scientific integrity, democratic responsibility, and human dignity.

With this position, the work concludes not as a catalogue of open problems but as a framework for thought—one that does not promise certainty, but offers clarity, distinction, and responsibility in a world that remains, inevitably, open.

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